Experiment 7

PL/SQL Procedures and Functions
Oracle Database 21c XE

1. Source Code

```
-- ------------------------------------------------------------
-- 7a. Stored procedure with IN and OUT parameters
-- ------------------------------------------------------------

-- 7a-1. Create the student table and insert the sample data
DROP TABLE student PURGE;
CREATE TABLE student (
  student_id   NUMBER(4) PRIMARY KEY,
  student_name VARCHAR2(30),
  course       VARCHAR2(20),
  marks        NUMBER(3)
);
INSERT INTO student VALUES (101,'Rahul','B.Tech',82);
INSERT INTO student VALUES (102,'Sneha','B.Tech',74);
INSERT INTO student VALUES (103,'Arjun','BCA',58);
INSERT INTO student VALUES (104,'Priya','B.Sc',91);
INSERT INTO student VALUES (105,'Kiran','BCA',45);
COMMIT;

-- 7a-2. Create the procedure GET_STUDENT_DETAILS
CREATE OR REPLACE PROCEDURE get_student_details (
  p_id    IN  NUMBER,
  p_name  OUT VARCHAR2,
  p_marks OUT NUMBER
) IS
BEGIN
  SELECT student_name, marks
  INTO   p_name, p_marks
  FROM   student
  WHERE  student_id = p_id;
EXCEPTION
  WHEN NO_DATA_FOUND THEN
    p_name  := NULL;
    p_marks := NULL;
    DBMS_OUTPUT.PUT_LINE('Student Not Found');
END;
/

-- 7a-3. Call the procedure from an anonymous block (ID 101)
SET SERVEROUTPUT ON
DECLARE
  v_name  VARCHAR2(30);
  v_marks NUMBER;
BEGIN
  get_student_details(101, v_name, v_marks);
  IF v_name IS NOT NULL THEN
    DBMS_OUTPUT.PUT_LINE('Student Name : ' || v_name);
    DBMS_OUTPUT.PUT_LINE('Marks        : ' || v_marks);
  END IF;
END;
/

-- 7a-4. Call the procedure with an ID that does not exist (ID 999)
-- same block with get_student_details(999, v_name, v_marks);

-- ------------------------------------------------------------
-- 7b. Stored functions
-- ------------------------------------------------------------

-- 7b-1. Program 1: create the employee table and insert the data
DROP TABLE employee PURGE;
CREATE TABLE employee (
  emp_id         NUMBER(4) PRIMARY KEY,
  emp_name       VARCHAR2(30),
  monthly_salary NUMBER(8,2)
);
INSERT INTO employee VALUES (1,'Ravi',25000);
INSERT INTO employee VALUES (2,'Anita',32000);
```

```
INSERT INTO employee VALUES (3,'Kiran',28500);
INSERT INTO employee VALUES (4,'Priya',40000);
COMMIT;

-- 7b-2. Create the function CALCULATE_ANNUAL_SALARY
CREATE OR REPLACE FUNCTION calculate_annual_salary (
  p_monthly_salary IN NUMBER
) RETURN NUMBER IS
BEGIN
  RETURN p_monthly_salary * 12;
END;
/

-- 7b-3. Invoke the function in a SELECT statement
SELECT emp_id, emp_name, monthly_salary,
       calculate_annual_salary(monthly_salary) AS annual_salary
FROM employee;

-- 7b-4. Program 2: create the student table and insert the data
DROP TABLE student PURGE;
CREATE TABLE student (
  student_id   NUMBER(4) PRIMARY KEY,
  student_name VARCHAR2(30),
  course       VARCHAR2(20),
  marks        NUMBER(3)
);
INSERT INTO student VALUES (101,'Rahul','B.Tech',82);
INSERT INTO student VALUES (102,'Sneha','B.Tech',74);
INSERT INTO student VALUES (103,'Arjun','BCA',58);
INSERT INTO student VALUES (104,'Priya','B.Sc',91);
INSERT INTO student VALUES (105,'Kiran','BCA',45);
COMMIT;

-- 7b-5. Create the function COUNT_STUDENTS
CREATE OR REPLACE FUNCTION count_students (
  p_course IN VARCHAR2
) RETURN NUMBER IS
  v_count NUMBER;
BEGIN
  SELECT COUNT(*)
  INTO   v_count
  FROM   student
  WHERE  course = p_course;
  RETURN v_count;
END;
/

-- 7b-6. Invoke the function in a SELECT statement
SELECT 'B.Tech' AS course,
       count_students('B.Tech') AS total_students
FROM dual;

-- 7b-7. Total students for every course
SELECT DISTINCT course,
       count_students(course) AS total_students
FROM student
ORDER BY course;

-- 7b-8. Program 3: create the student table (ID, name, marks)
DROP TABLE student PURGE;
CREATE TABLE student (
  student_id   NUMBER(4) PRIMARY KEY,
  student_name VARCHAR2(30),
  marks        NUMBER(3)
);
INSERT INTO student VALUES (101,'Rahul',82);
INSERT INTO student VALUES (102,'Sneha',74);
INSERT INTO student VALUES (103,'Arjun',58);
INSERT INTO student VALUES (104,'Priya',91);
INSERT INTO student VALUES (105,'Kiran',45);
INSERT INTO student VALUES (106,'Meena',30);
COMMIT;

-- 7b-9. Create the function GET_GRADE
CREATE OR REPLACE FUNCTION get_grade (
  p_marks IN NUMBER
) RETURN VARCHAR2 IS
  v_grade VARCHAR2(20);
```

```
BEGIN
  IF p_marks >= 75 THEN
    v_grade := 'Distinction';
  ELSIF p_marks >= 60 THEN
    v_grade := 'First Class';
  ELSIF p_marks >= 50 THEN
    v_grade := 'Second Class';
  ELSIF p_marks >= 35 THEN
    v_grade := 'Pass';
  ELSE
    v_grade := 'Fail';
  END IF;
  RETURN v_grade;
END;
/

-- 7b-10. Invoke the function in a SELECT statement
SELECT student_name, marks,
       get_grade(marks) AS grade
FROM student;
```

2. Execution and Output

Output shown is sample output of the programs above (Oracle Database 21c XE, SQL*Plus).

7a. Stored procedure with IN and OUT parameters

```
[7a-1]
Table dropped.

Table created.

1 row created.
1 row created.
1 row created.
1 row created.
1 row created.

Commit complete.

[7a-2]

Procedure created.

[7a-3]
Student Name : Rahul
Marks        : 82

PL/SQL procedure successfully completed.

[7a-4]
Student Not Found

PL/SQL procedure successfully completed.
```

7b. Stored functions

```
[7b-1]
Table dropped.

Table created.

1 row created.
1 row created.
1 row created.
1 row created.

Commit complete.

[7b-2]

Function created.

[7b-3]

EMP_ID EMP_NAME MONTHLY_SALARY ANNUAL_SALARY
------ -------- -------------- -------------
     1 Ravi              25000        300000
     2 Anita             32000        384000
     3 Kiran             28500        342000
     4 Priya             40000        480000

4 rows selected.

[7b-4]
Table dropped.

Table created.

1 row created.
1 row created.
1 row created.
1 row created.
1 row created.

Commit complete.

[7b-5]

Function created.

[7b-6]
```

```
COURSE TOTAL_STUDENTS
------ --------------
B.Tech              2

1 row selected.

[7b-7]

COURSE TOTAL_STUDENTS
------ --------------
B.Sc                1
B.Tech              2
BCA                 2

3 rows selected.

[7b-8]
Table dropped.

Table created.

1 row created.
1 row created.
1 row created.
1 row created.
1 row created.
1 row created.

Commit complete.

[7b-9]

Function created.

[7b-10]

STUDENT_NAME MARKS GRADE
------------ ----- ------------
Rahul           82 Distinction
Sneha           74 First Class
Arjun           58 Second Class
Priya           91 Distinction
Kiran           45 Pass
Meena           30 Fail

6 rows selected.
```